

1 L2–3 Supervised Learning

1.1 Classification and Loss

ERM: population risk $\theta^* = \arg \min_{\theta} \mathbb{E}_{x \sim p} \text{err}(f(x; \theta), y(x))$: empirical risk $\hat{\theta} = \arg \min_{\theta} \frac{1}{N} \sum \text{err}(f(x_i; \theta), y_i)$. **Binary logistic**: $p(y = 1|x) = \sigma(w^\top x + b)$,

$$\ell = -y \log p - (1 - y) \log(1 - p), \quad \frac{\partial \ell}{\partial z} = p - y.$$

Multi-class: logits $z \in \mathbb{R}^K$, $p_k = e^{z_k} / \sum_j e^{z_j}$,

$$\ell(z, y) = -\log p_y = -z_y + \log \sum_j e^{z_j}, \quad \frac{\partial \ell}{\partial z_k} = p_k - \mathbf{1}[k = y].$$

One-hot CE: $-\sum_k y_k \log p_k$.

1.2 MLP and Backprop

Layer: $h^{(0)} = x$, $z^{(\ell)} = W^{(\ell)} h^{(\ell-1)} + b^{(\ell)}$, $h^{(\ell)} = \phi(z^{(\ell)})$. Batch code convention often uses $X \in \mathbb{R}^{B \times n}$, $Z = XW + b$, $W \in \mathbb{R}^{n \times m}$. **Backprop**: define $\delta^{(\ell)} = \partial L / \partial z^{(\ell)}$: $\partial_{W^{(\ell)}} L = \delta^{(\ell)} (h^{(\ell-1)})^\top$, $\partial_{b^{(\ell)}} L = \delta^{(\ell)}$, $\delta^{(\ell-1)} = ((W^{(\ell)})^\top \delta^{(\ell)}) \odot \phi'(z^{(\ell-1)})$. **Activation**: sigmoid/tanh saturate; ReLU $\max(0, z)$ keeps positive-side gradient; ELU/smooth variants improve negative side. **Subgradient**: ReLU at 0 uses any value in $[0, 1]$ by convention.

1.3 CNN

Motivation: locality + weight sharing. Convolution:

$$y_{i,j,o} = \sum_{u,v,c} K_{u,v,c,o} x_{i+u,j+v,c} + b_o.$$

For input $H \times W \times C_{\text{in}}$, kernel $K_H \times K_W$, padding P , stride S : $H' = \lfloor (H + 2P - K_H) / S \rfloor + 1$, $W' = \lfloor (W + 2P - K_W) / S \rfloor + 1$. Params $K_H K_W C_{\text{in}} C_{\text{out}}$ + C_{out} . **Scanning MLP**: same detector slides across space. **Pooling**: max/avg reduces spatial size and jitter. **Padding**: keeps boundary info; same padding for odd kernel uses $P = (K - 1) / 2$.

1.4 Optimization

GD: $x_{k+1} = x_k - \eta_k \nabla f(x_k)$. **L-smooth**: $\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|$; if C^2 , $\nabla^2 f(x) \leq LI$. It implies $f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2$, so $0 < \eta < 2/L$ decreases the upper bound. **Strong convexity**: $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|y - x\|^2$; with L-smooth and C^2 , $\mu I \leq \nabla^2 f(x) \leq LI$. GD has linear convergence $k = O(\kappa \log(1/\epsilon))$, $\kappa = L/\mu$.

Newton: from quadratic Taylor, $x_{k+1} = x_k - \eta [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$.

$g_k = \text{gradient on } x_k$. **Momentum**: $\Delta X^k = \beta \Delta X^{k-1} - \eta \nabla f(X^k)$, $X^{k+1} = X^k + \Delta X^k$.

Nesterov: compute gradient at lookahead $x_k + \beta(x_k - x_{k-1})$.

SGD / mini-batch: use one or a small batch of samples to estimate the full gradient; mini-batch reduces variance while keeping updates cheap. **Adaptive LR**: per-coordinate step sizes help convergence. **AdaGrad**: $s_k = s_{k-1} + s_k^2$, $x_{k+1} = x_k - \eta s_k / (\sqrt{s_k} + \epsilon)$; rarely updated dimensions have smaller $s_{k,i}$ and larger effective learning rate, but learning can decay too fast.

RMSProp: $s_k = \rho s_{k-1} + (1 - \rho) s_k^2$.

Adam: $m_k = \beta_1 m_{k-1} + (1 - \beta_1) s_k$, $v_k = \beta_2 v_{k-1} + (1 - \beta_2) s_k^2$,

$x_{k+1} = x_k - \eta m_k / (\sqrt{v_k} + \epsilon)$, $\hat{m}_k = m_k / (1 - \beta_k^1)$.

1.5 Regularization and Architectures

Weight decay: L2 regularization stabilizes weights, $L_{\text{reg}}(w) = L_{\text{loss}}(w) + \alpha \|w\|_2^2$. **Early stopping**: stop when training improves but held-out validation worsens. **Dropout**: randomly zero units, prevents overfitting to particular neurons/weights, approximates ensemble and reduces co-adaptation; can be viewed as implicit L2. Two conventions: scale activations by keep-prob during training and leave inference unchanged, or leave training unchanged and multiply by keep-prob at inference. **Gradient clipping**: $g \leftarrow g \min(1, \tau / \|g\|)$. **Normalization**: **GroupNorm**: batch-independent. **InstanceNorm**: batch-independent, suitable for generation tasks. **BatchNorm**: $m_k = \frac{1}{B} \sum_i z_i$, $\sigma_B^2 = \frac{1}{B} \sum_i (z_i - \mu_B)^2$, $z_i = \gamma \frac{z_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} + \beta$. Inference uses running averages because there is no mini-batch training distribution: $\mu_{\text{run}} \approx \frac{1}{N_{\text{batch}}} \sum_{\text{batch}} \mu_B$, $\sigma_{\text{run}}^2 \approx \frac{1}{N_{\text{batch}}} \sum_{\text{batch}} \frac{1}{B-1} \sigma_B^2$. Backprop through a layer uses Jacobian transpose: $\nabla_x L = J^\top \nabla_y L$. **PCA color jitter**: $[I_R, I_G, I_B] \leftarrow [I_R, I_G, I_B] + \alpha [P_R, P_G, P_B] [\lambda_R, \lambda_G, \lambda_B]^\top$, $\alpha \sim \mathcal{N}(0, 0.1)$; perturb RGB along principal color directions. **LayerNorm**: normalize hidden dimension inside one sample; better for sequence models. Batch-independent, Particularly suitable for RNN. $h^t = f\left(\frac{g}{\sigma} \odot (a^t - \mu^t) + b\right)$, $\mu^t = \frac{1}{H} \sum_i a_i^t \sigma_i^t = \sqrt{\frac{1}{H} \sum_i (a_i^t - \mu^t)^2}$.

ResNet: $h_{t+1} = h_t + F(h_t)$, residual path eases optimization. **DenseNet**: concatenate previous features. **FCN**: replace fully connected layer by convolution for dense prediction.

2 L4 Energy-Based Models

2.1 Energy View

Generative modeling: learn $P(X)$ or $P(X, Y) = P(Y)P(X|Y)$ instead of $P(y|X)$. Generative models can be used to both classify and generate images **EBM**: unnormalized score via energy

$$p_{\theta}(x) = \frac{e^{-E_{\theta}(x)}}{Z_{\theta}}, \quad Z_{\theta} = \int e^{-E_{\theta}(x)} dx.$$

Low energy means high probability; training and inference both need sampling or partition-function gradients. $\log p_{\theta}(x) = -E_{\theta}(x) - \log Z_{\theta}$, $\nabla_{\theta} \log p_{\theta}(x) = -\nabla_{\theta} E_{\theta}(x) - \frac{1}{Z_{\theta}} \nabla_{\theta} \left(\sum_{x'} e^{-E_{\theta}(x')} \right) = -\nabla_{\theta} E_{\theta}(x) + \mathbb{E}_{x' \sim p_{\theta}} [E_{\theta}(x')]$.

We can train an energy-based model without knowing the explicit density function (or normalizing factor Z). Such as score-matching, etc.

2.2 Hopfield Network

State $y_i \in \{-1, 1\}$, local field $h_i = \sum_{j \neq i} w_{ij} y_j + b_i$. Update flips if $y_i h_i < 0$.

$$E(y) = -\sum_{i < j} w_{ij} y_i y_j - \sum_i b_i y_i = -\frac{1}{2} y^\top W y - b^\top y.$$

Asynchronous flip changes energy by $\Delta E = 2y_i h_i$; if $y_i h_i < 0$, energy decreases. Finite states imply convergence to a local minimum. **Hebbian storage**: $w_{ij} = \frac{1}{N^2} \sum_p y_i^p y_j^p$, $w_{ii} = 0$. For N neurons, $K \leq N / (4 \log N)$ random patterns can be stable. Multiple patterns cause **spurious minima**; capacity is limited.

Objective for W : $W = \arg \min_W \left(\sum_{y \in P} E(y) - \sum_{y' \notin P} E(y') \right)$, $E(y) = -\frac{1}{2} y^\top W y$. **Optimization view**: desired patterns should be valleys; short-run evolution can raise undesired valleys by contrastive updates.

SGD update rule for W : $W \leftarrow W + \eta \left(\mathbb{E}_{y \in P} [y y^\top] - \mathbb{E}_{y' \notin P} [y' y'^\top] \right)$; in practice sample desired y , initialize y' from a desired pattern, run 2–4 timesteps, update with $y y^\top - y' y'^\top$.

2.3 Boltzmann Machine

Stochastic Hopfield: energy $E(y)$ as Hopfield, $p(y) = \frac{1}{Z} e^{-E(y)} \propto e^{-E(y)}$. Conditional Gibbs update for $\{-1, 1\}$ units: for $h_i = \sum_j w_{ij} y_j + b_i$,

$$P(y_i = 1|y_{-i}) = \frac{1}{1 + e^{-2h_i}} = \sigma(2h_i), \quad h_i = \sum_j w_{ij} y_j + b_i.$$

$$\log \frac{P(y_i = 1|y_{-i})}{1 - P(y_i = 1|y_{-i})} = 2h_i.$$

Stochastic evolution: initialize $y^{(0)}$, sample coordinates $y_i(t+1) \sim \text{Bernoulli}(\sigma(z_i(t)))$, repeat; recover pattern by last- M majority $\hat{y}_i = \frac{1}{M} \sum_{t=L-M+1}^T y_i(t) > 0$. Temperature T : $y_i(T) \sim \text{Bernoulli}(\sigma(z_i(T)/T))$, $T \leftarrow \alpha T$; annealing lowers T from exploration to exploitation.

BM Training: maximize $L(W) = \sum_{y \in P} \log P_{\theta}(y)$, θ means parameters.

$$P(y) = \frac{\exp(\frac{1}{2} y^\top W y)}{\sum_{y'} \exp(\frac{1}{2} y'^\top W y)}, \quad L(W) = \frac{1}{N_P} \sum_{y \in P} \frac{1}{2} y^\top W y - \log \sum_{y'} e^{\frac{1}{2} y'^\top W y}.$$

$\nabla w_{ij} \log p_{\theta}(y) = y_i y_j - \mathbb{E}_{p_{\theta}} [y_i y_j]$, $\nabla w_{ij} L = \frac{1}{N_P} \sum_{y \in P} y_i y_j - \frac{1}{M} \sum_{y' \in S} y'_i y'_j$. First term: compute average over P . Second term: sample from p_{θ} , compute average. **Sample method**: random initialize $y^{(0)}$, cycle over i (coordinates) and sample from $P(y_i(t)|y_{-i}(t))$, after convergence select the final M states as samples. **Updating**: $w_{ij} \leftarrow w_{ij} + \eta \nabla w_{ij} L(W)$. Learning is **positive phase** data correlation minus **negative phase** model correlation. **With hidden neurons**: $y = (v, h)$

$$\nabla w_{ij} L = \frac{1}{|P|} \sum_{v \in P} \mathbb{E}_{h|v} [v_i y_j] - \mathbb{E}_{y' \sim p_W} [y'_i y'_j].$$

Use conditioned samples S_c by clamping visible v and Gibbs-sampling hidden h , and free samples S by running all variables: $\nabla w_{ij} L = \frac{1}{N_P K} \sum_{y \in S_c} y_i y_j - \frac{1}{M} \sum_{y' \in S} y'_i y'_j$.

Boltzmann for classification: $y = (v, h, c)$, c : visible, one-hot vector

2.4 RBM and Contrastive Divergence

RBM: bipartite v, h ; no visible-visible or hidden-hidden edges: $E(v, h) = -v^\top W h - b^\top v - c^\top h$. Conditionals factorize: $P(h_j = 1|v) = \sigma(c_j + \sum_i w_{ij} v_i)$, $P(v_i = 1|h) = \sigma(b_i + \sum_j w_{ij} h_j)$, $\sigma(z) = \frac{1}{1 + e^{-z}}$. Sampling: $v^{(0)} \rightarrow h^{(0)} \rightarrow v^{(1)} \rightarrow h^{(1)} \rightarrow \dots$

Maximum likelihood estimate: $\nabla w_{ij} L = \frac{1}{N_P K} \sum_{v \in P} v_i^0 h_j^0 - \frac{1}{M} \sum v_i^\infty h_j^\infty$. CD-1 approximation:

$$\nabla w_{ij} L \approx \frac{1}{N_P} \sum_{v \in P} (v_i^0 h_j^0 - v_i^1 h_j^1).$$

CD-k: start at data $v^{(0)}$, run k Gibbs steps, update $W \propto v^{(0)} h^{(0)\top} - v^{(k)} h^{(k)\top}$.

2.5 Sampling

Inverse CDF: sample $u \sim U(0, 1)$, choose category by cumulative mass. **Box-Muller**: Gaussian from uniform:

$$z_1 = \sqrt{-2 \log u_1} \cos(2\pi u_2), \quad z_2 = \sqrt{-2 \log u_1} \sin(2\pi u_2).$$

Importance sampling: $\mathbb{E}_P[f(x)] = \mathbb{E}_Q[f(x)p(x)/q(x)]$. Good q must cover p 's high-mass region. **MCMC**: **Detailed balance** $\pi(x)T(x \rightarrow x') = \pi(x')T(x' \rightarrow x)$ implies stationarity of π . **Ergodic**: can visit any desired state with positive probability and has a unique stationary distribution; a strong minorization form is $\min_{z, z': \pi(z') > 0} P(z \rightarrow z') / \pi(z') = \delta > 0$. **Metropolis Hastings Algo.**: Construct Markov chain with stationary π (desired distribution). Current x , draw $x' \sim q(x'|x)$ (called proposal), transition to x' with prob. $\alpha = \min\left(1, \frac{\pi(x')q(x|x')}{\pi(x)q(x'|x)}\right)$, otherwise stays at x . Choice of q : random proposal $q(x'|x) = \text{noise}$ (e.g. Gaussian, then it relies on $|x - x'|$), then q terms in α vanishes. **For EBM, partition Z cancels**.

Gibbs: sample a single coordinate per step from $q(s_i \rightarrow s'_i) = p(s'_i | s_{j \neq i})$; it is MH with acceptance probability 1.

SGLD: $x_{t+1} = x_t - \frac{\eta}{2} \nabla_x E_{\theta}(x_t) + \sqrt{\eta} \xi_t$.

3 L5 Generative Adversarial Networks

3.1 Implicit Generative Model

Instead of density $p_{\theta}(x)$, learn sampler $x = G_{\theta}(z)$, $z \sim p_z$. Likelihood-free learning. Need differentiable distance between generated and real distributions. GAN learns this distance with a discriminator. **GAN**: generator $G_{\theta}(z)$, $z \sim p(z) = N(0, I)$; discriminator $D_{\phi}(x)$, classify the data is from p_{data} or G_{θ} . Objective: $L = \min_{\theta} \max_{\phi} V(D_{\phi}, G_{\theta})$,

$$V(D_{\phi}, G_{\theta}) = \mathbb{E}_{x \sim p_{\text{data}}} \log D_{\phi}(x) + \mathbb{E}_{z \sim p_z} \log(1 - D_{\phi}(G_{\theta}(z))).$$

Dataset: $\{(x, 1) : x \sim p_{\text{data}}\} \cup \{(\hat{x}, 0) : \hat{x} \sim G_{\theta}(z)\}$.

Train D : $L(\phi) = \mathbb{E}_{x \sim p_{\text{data}}} \log D_{\phi}(x) + \mathbb{E}_{z \sim p_z} \log(1 - D_{\phi}(G(z)))$.

Train G : $L(\theta) = \mathbb{E}_{z \sim p(z)} [\log D_{\phi}(G_{\theta}(z))]$.

For fixed G , the optimal D is $D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_G(x)}$. Substitute:

$$V(D^*, G) = -\log 4 + 2 \text{JSD}(p_{\text{data}} \| p_G), \quad \text{JSD}: \text{JSD}(p \| q) = \text{KL}(p \| \frac{p+q}{2}) + \text{KL}(q \| \frac{p+q}{2}), \geq 0, \sqrt{\text{JSD}}$$
 satisfies triangular inequality.

$$2\text{KL}(p \| q) \geq \int |p(x) - q(x)| dx^2.$$

Non-saturating generator loss $-\mathbb{E}_z \log D(G(z))$ gives stronger gradients.

3.2 Evaluation and Failure Modes

Mode collapse: generator covers few modes. For multimodal p , reverse KL $\text{KL}(q \| p) = \mathbb{E}_q \log(q/p)$ and JSD often prefer one high-quality mode; forward KL $\text{KL}(p \| q) = \mathbb{E}_p \log(p/q)$ penalizes missing modes and is mode-covering. JSD: $\frac{1}{2} \text{KL}(p \| m) + \frac{1}{2} \text{KL}(q \| m)$, $m = (p + q)/2$. **Training instability**: moving target game can oscillate; discriminator too strong/simple $\Rightarrow$ generator gradient vanishes. **Inception Score**:

$$\text{IS} = \exp\left(\mathbb{E}_{x \sim G} \text{KL}(f(y|x) \| p_f(y))\right), \quad p_f(y) = \mathbb{E}_{x \sim G} f(y|x).$$

Since $\mathbb{E}_x \text{KL}(f(\cdot|x) \| p_f) = -\mathbb{E}_x H(f(\cdot|x)) + H(p_f)$, high IS needs confident images and diverse classes. IS never uses p_{data} , so it cannot detect memorization. **FID**: Compute μ_P, Σ_P and μ_G, Σ_G for $p_{\text{data}}, G(z)$ using Inception-v3 pool3 layer (2048-d); compare Gaussian feature statistics: $\|\mu_P - \mu_G\|_2^2 + \text{tr}(\Sigma_P + \Sigma_G - 2(\Sigma_P \Sigma_G)^{1/2})$. Lower FID score means better generators.

3.3 GAN Training Techniques

Feature Matching: train G to match discriminator feature statistics:

$$L_G(\theta) = \left\| \mathbb{E}_{\hat{x} \sim G} f(\hat{x}; \phi) - \mathbb{E}_{x \sim p_{\text{data}}} f(x; \phi) \right\|_2^2.$$

where $f(x; \phi)$ is the final activation layer of D . **Other tricks**: One-Sided Label Smoothing; Historical Averaging; Minibatch Discrimination; Virtual Batch Normalization.

3.4 WGAN Family

JSD can be flat when supports do not overlap. Wasserstein-1: $W(p, q) = \sup \|f\|_{L \leq 1} \mathbb{E}_{x \sim p} f(x) - \mathbb{E}_{x \sim q} f(x)$.

WGAN: critic f_{ϕ} , no D . WGAN objective:

$$\min_G \max_{D: \|D\|_{L \leq 1}} \mathbb{E}_{x \sim p_{\text{data}}} [D(x)] - \mathbb{E}_{\hat{x} \sim p_G} [D(\hat{x})].$$

Equivalently maximize critic $L(\phi) = \mathbb{E}_{x \sim p_{\text{data}}} f_{\phi}(x) - \mathbb{E}_{\hat{x} \sim p_G} f_{\phi}(\hat{x})$ and train G to reduce the critic gap.

Lipschitz constraint: weight clipping, spectral norm, or **WGAN-GP**: $\hat{x} \leftarrow (1 - \epsilon)x + \epsilon \hat{x}$, $\epsilon \sim \text{Unif}(0, 1)$, $L(\phi) = \mathbb{E}_{x \sim p_{\text{data}}} [f_{\phi}(x)] - \mathbb{E}_{\hat{x} \sim p_G} [f_{\phi}(\hat{x})] + \lambda \mathbb{E}_{\hat{x}} (\|\nabla f_{\phi}(\hat{x})\|_2 - 1)^2$.

DCGAN: conv/deconv, batchnorm, stable architecture. **BigGAN**: scale + class conditioning + truncation trick. **GigaGAN**: text-to-image GAN scaling. **R3GAN**: regularized robust objective.

3.5 Variants and Applications

Conditional GAN: $G(z, c)$, $D(x, c)$. **Semi-supervised GAN**: discriminator has $K + 1$ classes, K real labels plus fake. **BiGAN/ALI**: learn encoder $E(x)$ with discriminator over (x, z) pairs. Idea: $(z, G(z))$ and $(E(x), x)$ from same distribution. gives representation learning. **Pix2Pix**: paired image-to-image translation with conditional GAN + reconstruction loss. **CycleGAN**: unpaired translation using cycle consistency $G: X \rightarrow Y$, $F: Y \rightarrow X$, $\|F(G(x)) - x\| + \|G(F(y)) - y\|$. **DragGAN**: interactive image manipulation by optimizing latent/code with handle-point constraints.

4 L6 Flow and VAE

4.1 Normalizing Flow

Invertible differentiable generator $x = G_{\theta}(z)$, transform a simple distribution to the data distribution. $\log p_X(x) = \log p_Z(G^{-1}(x)) + \log \left| \det \frac{\partial G^{-1}}{\partial x} \right|$.

Composition: $z_K = G(z_0) = f_K \circ \dots \circ f_1(z_0)$, $\log p_X(z_K) = \log p_Z(z_0) - \sum_k \log |\det \frac{\partial f_k}{\partial z_k}|$. Flow design needs invertible layer, edge inverse, efficient compute determinant of Jacobi.

Coupling: split $x = (x_a, x_b)$, to (y_a, y_b) , $y_a = x_a$, $y_b = x_b \odot e^{s(x_a)} + t(x_a)$, $\log |\det J| = \sum s(x_a)$. s, t : learnable. Sometimes needs permutation among coordinates or linear transformation.

Glow: invertible 1×1 convolution mixes channels. Since the same channel-mixing matrix W is applied at each spatial location,

$$\log \left| \det \frac{d \text{conv}2D(h; W)}{dh} \right| = H W_{\text{spatial}} \log |\det W|.$$

LU parameterization $W = PLU + \text{diag}(s)$ makes determinant cheap: $\log |\det W| = \sum_i \log |s_i|$. **Dequantization**: add uniform noise to discrete pixels so continuous density is well-defined. **Continuous NF / Neural ODE**: $\frac{dz(t)}{dt} = f_{\theta}(z(t), t)$, $\log p(z(t_1)) = \log p(z(t_0)) - \int_{t_0}^{t_1} \text{tr} \left(\frac{\partial f_{\theta}}{\partial z(t)} \right) dt$.

Prove a Normalizing Flow: 1. invertible; 2. easily-tractable Jacobi matrix,

4.2 GMM and EM

GMM responsibility: for K mixture components,

$$\gamma_{ij} := p(z_i = j | x_i) = \frac{\pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}{\sum_{\ell=1}^K \pi_{\ell} \mathcal{N}(x_i | \mu_{\ell}, \Sigma_{\ell})}.$$

GMM EM lower bound:

$$\sum_i \log \sum_j \gamma_{ij} \frac{\pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}{\gamma_{ij}} \geq \sum_i \sum_j \gamma_{ij} \log \frac{\pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}{\gamma_{ij}}.$$

E-step fixes parameters and maximizes over γ :

$$\gamma^{(t+1)} = \arg \max_{\gamma_{ij} \geq 0, \sum_j \gamma_{ij} = 1} \sum_j \gamma_{ij} \log \pi_j^{(t)} \mathcal{N}(x_i | \mu_j^{(t)}, \Sigma_j^{(t)}) - \sum_{i,j} \gamma_{ij} \log \gamma_{ij}.$$

M-step fixes γ and maximizes

$$(\pi, \mu, \Sigma)^{(t+1)} = \arg \max_{\pi, \mu, \Sigma} \sum_{i,j} \gamma_{ij}^{(t+1)} \log \pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j).$$

4.3 Evidence Lower Bound (ELBO)

Latent-variable likelihood $\log p_{\theta}(x) = \log \sum_z p_{\theta}(x, z)$. Jensen:

$$\log p_{\theta}(x) = \log \mathbb{E}_q \frac{p_{\theta}(x, z)}{q(z)} \geq \mathbb{E}_q \log p_{\theta}(x, z) + H(q) = \mathcal{L}(q, \theta).$$

Gap: $\log p_{\theta}(x) - \mathcal{L}(q, \theta) = \text{KL}(q(z) \| p_{\theta}(z|x))$. Equivalently

$$\log p_{\theta}(x) \geq \int q_{\theta}(z|x) \log p_{\theta}(x|z) dz - \text{KL}(q_{\theta}(z|x) \| p(z)),$$

with gap $\text{KL}(q_{\theta}(z|x) \| p_{\theta}(z|x))$. EM is coordinate ascent on this lower bound: E-step sets $q(z) = p_{\theta \text{old}}(z|x)$, M-step maximizes $\mathbb{E}_q \log p_{\theta}(x|z)$.

4.4 VAE

Encoder $q_X(z) = \mathcal{N}(\mu_{\theta}(x), \sigma_{\theta}(x))$, decoder $p_{\psi}(x|z) = \mathcal{N}(\psi(z), \beta I)$, prior $p(z)$. ELBO: $\log p(x) \geq \int q_X(z) \log p_{\psi}(x|z) dz - \text{KL}(q_X(z) \| p(z))$. With Gaussian decoder,

$$\text{ELBO} = -\frac{1}{2\beta} \mathbb{E}_{q_X(z)} \|\psi(z) - x\|_2^2 - \text{KL}(q_X(z) \| p(z)) + C.$$

Reparameterization: $z = \mu_{\theta}(x) + \sigma_{\theta}(x)\epsilon$, $\epsilon \sim \mathcal{N}(0, I)$. **Gaussian KL**: for diagonal Gaussians $p = \mathcal{N}(\mu_1, \sigma_1^2)$, $q = \mathcal{N}(\mu_2, \sigma_2^2)$,

$$\text{KL}(p\|q) = \log \frac{\sigma_2^2}{\sigma_1^2} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

For $p(z) = \mathcal{N}(0, I)$, $\text{KL}(q_X(z) \| p(z)) = \frac{1}{2} \|\mu_{\theta}(x)\|^2 + \frac{1}{2} \sigma_{\theta}^2(x) - \log \sigma_{\theta}(x) + C$.

Terms: reconstruction = data fidelity; KL = latent regularization. In β -VAE, large β enforces latent variables to be less correlated with each other:

$$\mathcal{L}_{\beta\text{-VAE}} = \mathbb{E}_{q_{\theta}(z|x)} [\log p_{\theta}(x|z)] - \beta \text{KL}(q_{\theta}(z|x) \| p(z)).$$

PixelVAE/NVAE: hierarchical latent variables improve global coherence and likelihood.

5 L7 Diffusion and Flow Matching

5.1 DDPM

Forward: gradually add Gaussian noise,

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I), \quad \alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I), \quad x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon.$$

Reverse: learn denoising transitions,

$$p(x_{0:T}) = p(x_T) \prod_{t=1}^T p(x_{t-1}|x_t), \quad p(x_T) = \mathcal{N}(0, I),$$

$$p(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_{\theta}(x_t, t), \sigma_t^2 I).$$

Loss / ELBO identity: reuse the Gaussian-product identity

$$\log q(x_t|x_{t-1}) + \log q(x_{t-1}|x_0) = \log q(x_{t-1}|x_t, x_0) + \log q(x_t|x_0).$$

The generic VAE-style decomposition is

$$-\log p(x) = -\int q_{\theta}(z) \log p(x|z) dz + \text{KL}(q_{\theta}(z) \| p(z)) - \text{KL}(q_{\theta}(z) \| p(z|x)).$$

True posterior:

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t I), \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t,$$

$$\tilde{\mu}_t = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t.$$

Gaussian KL reduces each timestep to mean matching:

$$L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(x_t, x_0) - \mu_{\theta}(x_t, t)\|^2 \right] + C.$$

Noise prediction: since

$$\tilde{\mu}_t = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right), \quad \mu_{\theta} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right),$$

$$L_{t-1} = \mathbb{E}_{x_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \bar{\alpha}_t (1 - \bar{\alpha}_t)} \|\epsilon - \epsilon_{\theta}(x_t, t)\|^2 \right] + C.$$

Training input is (x_t, t) and target is noise ϵ ; $\epsilon_{\theta}(x_t, t)$ is usually a U-Net with time encoding. **Sampling** from $x_T \sim \mathcal{N}(0, I)$:

$$x_{t-1} = \frac{1}{\sqrt{\bar{\alpha}_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z, \quad z \sim \mathcal{N}(0, I).$$

5.2 Score-Based Models

Score: learn score $\nabla_x \log p(x)$. **Langevin dynamics**: use only score. $x_{t+1} = x_t + \eta \nabla_x \log p(x_t) + \sqrt{2\eta} \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, I)$. continuous form: $dx_t = \nabla_x \log p(x_t) dt + \sqrt{2d} w_t$, $dw \sim \mathcal{N}(0, dt)$. **NCSN**: noise conditional score network. $x_t = x_0 + \sigma_t \epsilon$, learn $s_{\theta}(x_t, \sigma_t)$ to estimate $\nabla_x \log p(x_t)$.

$x_t = a_t x_0 + \sigma_t \epsilon$, score $s_{\theta}(x_t, t)$? MSE loss $L = \mathbb{E}_{x_t} \|\bar{s}_{\theta}(x_t, t) - \nabla_{x_t} \log p(x_t)\|^2$. $\mathbb{E}_{x_t} [s_{\theta}(x_t, t)^{\top} \nabla_{x_t} \log p(x_t)] = \mathbb{E}_{x_t} [\nabla \cdot s_{\theta}(x_t, t)]$, using $\nabla \cdot (sp) = (\nabla \cdot s)p + s^{\top} \cdot \nabla p$. **Denoising score matching**: $\nabla_{x_t} \log p(x_t) = \mathbb{E}_{x_0 \sim p(\cdot|x_t)} [\nabla_{x_t} \log p(x_t|x_0)]$. $\mathbb{E}_{x_t} \|s_{\theta}(x_t, t) - \nabla_{x_t} \log p(x_t|x_0)\|^2$,

$$L_{DSM} = \mathbb{E}_{x_0, t, x_t} \|s_{\theta}(x_t, t) - \nabla_{x_t} \log q(x_t|x_0)\|^2,$$

where $x_t \sim q(x_t|x_0)$ is tractable. If $x_t = a_t x_0 + \sigma_t \epsilon$, then $\nabla_{x_t} \log q(x_t|x_0) = -\frac{x_t - a_t x_0}{\sigma_t^2}$. Let $s_{\theta}(x_t, t) = -\frac{\epsilon_{\theta}(x_t, t)}{\sigma_t}$; loss matches DDPM.

SDE: $dx = f(x, t)dt + g(t)dW_t$, reverse: $dx = [f(x, t) - g(t)^2 \nabla_x \log p_t(x)]dt + g(t)d\bar{W}_t$. A equivalent ODE (same distribution when same t): Probability-flow ODE: $dx = \left[f(x, t) - \frac{1}{2} g(t)^2 \nabla_x \log p_t(x) \right] dt$. Likelihood along ODE: $\log p_0(x_0) = \log p_T(x_T) + \int_0^T \text{tr} \left(\frac{\partial g}{\partial x_t} \right) dt$.

Inference: use learned score to solve SDE/ODE using a solver. Hence #sample-steps in training and inference can be different. DDIM: sample according to schedule

5.3 DPM-Solver and Applications

DPM solver 1 order DPM = DDIM. DPM-Solver integrates diffusion ODE with high-order exponential integrators, reducing sampling steps. **Latent diffusion**: diffuse in autoencoder latent space, cheaper than pixel space. **DiT/SORA**: transformer over spacetime latent patches. **DALL-E 2**: CLIP prior + diffusion decoder. **DALL-E 3**: synthetic captions improve prompt following. **SDEdit**: add noise to input image then denoise under prompt. **ControlNet**: extra condition branch injects edges/depth/pose. **Logit-normal sampling**: biases timestep sampling toward informative middle region.

5.4 Flow Matching

Continuous flow $dx_t/dt = v_{\theta}(x_t, t)$. Objective: $\mathcal{L}_{FM} = \mathbb{E}_{t, x_t} \|v_{\theta}(x_t, t) - u_t(x_t)\|^2$. Conditional path from noise x_0 to data x_1 : $x_t = (1-t)x_0 + tx_1$, target $u_t = x_1 - x_0$. Conditional Flow Matching (CFM) samples pairs and regresses conditional velocity; marginal vector field is recovered in expectation. **Non-uniqueness**: many vector fields induce same marginals; choice affects training variance and sampling trajectory. **SD3**: flow target often parameterizes velocity between noise/data under continuous timestep.

6 L8–9 Sequence Modeling

6.1 Sequence Data and Tokenization

Sequence data includes language, audio, action; images/DNA/protein can be serialized. Autoregressive factorization: $p(x_{1:T}) = \prod_{t=1}^T p(x_t|x_{<t})$. **Tokenization**: text $\rightarrow$ basic units $\rightarrow$ token IDs. Byte-Pair Encoding (BPE) repeatedly merges frequent adjacent symbols; balances vocabulary size and sequence length. **Embedding**: token ID $i \rightarrow e_i = W_E[i]$; positional info is needed because token embeddings alone are order-free.

6.2 Autoregressive CNN and Pixel Models

WaveNet: causal/dilated temporal convolution expands receptive field without recurrence. **PixelCNN**: image likelihood $p(x) = \prod_i p(x_i|x_{<i})$. under raster order; masked convolution prevents access to future pixels. Gated PixelCNN uses gates to improve expressiveness.

6.3 RNN and LSTM

RNN: $h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$, $y_t = W_{hy}h_t + b_y$. BPTT multiplies many Jacobians; singular values < 1 vanish, > 1 explode. **LSTM**: $f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$, $i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$, $\tilde{c}_t = \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c)$, $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$, $o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$, $h_t = o_t \odot \tanh(c_t)$. Additive cell path gives controllable memory; forget gate decides retention.

6.4 Attention and Transformer

For token t : $q_t = W_Q x_t$, $k_j = W_K x_j$, $v_j = W_V x_j$, $\alpha_{tj} = \frac{\exp(q_t^{\top} k_j / \sqrt{d_k})}{\sum_i \exp(q_t^{\top} k_i / \sqrt{d_k})}$, $y_t = \sum_j \alpha_{tj} v_j$. Matrix form: $\text{Attn}(Q, K, V) = \text{softmax}(QK^{\top} / \sqrt{d_k})V$. **Multi-head**: parallel attention subspaces, concatenate and project. **Causal mask**: block future tokens. **Transformer block**: attention + MLP + residual

+ normalization. Complexity $O(T^2 d)$; FlashAttention reduces memory IO. Mamba/SSM explores linear-time sequence mixing.

6.5 Objectives and Case Studies

Objectives: autoregressive LM, masked/denoising, contrastive predictive. **Recipe**: pretraining, fine-tuning, low-rank tuning. **GPT**: decoder-only causal Transformer; next-token pretraining gives in-context learning with scale. **ViT**: image patches as tokens. **Swin**: shifted local windows for hierarchical vision. **CLIP**: contrastive image-text alignment,

$$L = -\frac{1}{2} \left[\frac{1}{N} \sum_i \log \frac{e^{s_{ii}^{\text{img}/\tau}}}{\sum_j e^{s_{ij}^{\text{img}/\tau}}} + \frac{1}{N} \sum_i \log \frac{e^{s_{ii}^{\text{txt}/\tau}}}{\sum_j e^{s_{ji}^{\text{txt}/\tau}}} \right].$$

SigLIP: pairwise sigmoid loss avoids global softmax. **Whisper**: audio Transformer trained on large-scale speech. **MAR**: masked autoregressive modeling for images.

7 L10 Modern Large Models

7.1 Pretraining, Scaling, Prompting

Pipeline: Pretraining $\rightarrow$ Base GPT $\rightarrow$ SFT $\rightarrow$ RLHF $\rightarrow$ ChatGPT-like. Pre-training builds capability by next-token prediction; post-training reshapes behavior into instruction-following. Base GPT tends to continue text; chat model tries to answer the user. **Scaling laws**: held-out loss decreases smoothly as model size N , data D , and compute C grow, approximately by power law. The important lesson is bottleneck balance: scale parameters, tokens, and compute together. Useful heuristics: data should grow with model size ($D \propto N^{0.74}$); compute-optimal training spends extra compute mostly on larger N , somewhat on batch B , and barely on serial steps S . **Prompting**: zero-shot/few-shot puts task description/examples in context without parameter updates. GPT-3 made in-context learning practical; instruction tuning makes this behavior reliable by training on natural-language tasks.

7.2 SFT and RLHF

SFT: learn from human demonstrations (x, y) by next-token imitation:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{\text{D}_{\text{SFT}}} \sum_t \log \pi_{\theta}(y_t | x, y_{<t}).$$

It teaches desired response style but only uses positive examples, so it cannot directly rank two acceptable answers. **Reward model**: pairwise preference $y_w \succ y_l$ teaches scalar reward $r_{\phi}(x, y)$:

$$p_{\phi}(y_w \succ y_l | x) = \sigma(r_{\phi}(x, y_w) - r_{\phi}(x, y_l)).$$

$$\mathcal{L}_{\text{RM}} = -\mathbb{E} \log \sigma(r_{\phi}(x, y_w) - r_{\phi}(x, y_l)).$$

RLHF: optimize policy reward while staying near reference/SFT model:

$$\max_{\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x)} \left[r_{\phi}(x, y) - \beta \text{KL}(\pi_{\theta}(\cdot|x) \| \pi_{\text{ref}}(\cdot|x)) \right].$$

Why KL: prevents reward hacking and language-quality drift. **Pretrain vs post-train**: pretraining compresses web text distribution; post-training optimizes human utility/preference over responses.

7.3 Hallucination, RAG, Reasoning

Hallucination: SFT can teach a confident answer even when the base model lacks knowledge. Preference/RLHF can reward uncertainty awareness, e.g. prefer “I don’t know” over wrong entities. Negative samples (x, y^+, y^-) are valuable because they mark bad-but-plausible behavior. **RAG**: combine parametric memory in θ with retrieved non-parametric memory z :

$$p(y|x) \approx \sum_{z \in \text{TopK}(x)} p_{\eta}(z|x) p_{\theta}(y|x, z).$$

RAG helps latest knowledge without retraining, but retrieval is not the same as reasoning. **LRM**: reasoning model spends test-time compute on internal thinking before final answer:

$$x \rightarrow z_{1:m} \rightarrow y.$$

CoT prompting elicits intermediate reasoning through prompt examples; reasoning RL trains the behavior with verifier reward. Thinking labels are not unique, so RL can reward final correctness without prescribing one chain. Trajectory:

$$\tau = (z_1, \dots, z_m, y_1, \dots, y_T), \quad R(x, \tau) = \begin{cases} +1, & V(x, y) = \text{correct}, \\ -1, & V(x, y) = \text{incorrect}. \end{cases}$$

Agent loop: reasoning can extend to environment interaction with actions:

$$s_t = (x, \text{memory}, \text{observations}_{<t}), \quad a_t \in \{\text{search, browse, code, write, } \dots\}.$$

7.4 Multimodal Models

Grounding: multimodal models condition on text plus image/video/audio/interface inputs:

$$p_{\theta}(y|x_{\text{text}}, x_{\text{image}}, x_{\text{video}}, \dots).$$

Flamingo: extends GPT-style few-shot prompting to interleaved text-image/video input:

$$p_{\theta}(y|t_1, v_1, t_2, v_2, \dots, t_k, v_k, t_{k+1}).$$

Architecture: frozen vision encoder + frozen LM; Perceiver Resampler compresses visual features; gated cross-attention injects vision while protecting frozen LM. **LlaVA**: visual instruction tuning samples $(I, x_{\text{instruction}}, y_{\text{assistant}})$: CLIP features are projected into LM embedding space:

$$F_I = f_{\text{CLIP}}(I), \quad Z_I = F_I W, \quad [Z_I; E(x_1); \dots; E(x_T)].$$

Two-stage training: first align visual features by training projection W ; then freeze vision encoder and tune projection + LM for assistant behavior. **Unified multimodal/BAGEL**: one decoder for understanding/generation/editing/reasoning on interleaved data $t_1, t_1, t_2, t_2, \dots, v_1, t_k$; ViT pathway supports understanding, VAE pathway supports generation/reconstruction. Emergence order: understanding $\rightarrow$ generation $\rightarrow$ editing $\rightarrow$ reasoning.

8 Homework

Log-sum inequality: for $a_i, b_i \geq 0$ and positive sums,

$$\sum_i a_i \log \frac{a_i}{b_i} \geq \left(\sum_i a_i \right) \log \frac{\sum_i a_i}{\sum_i b_i}.$$

Two-term form:

$$(a_1 + a_2) \log \frac{a_1 + a_2}{b_1 + b_2} \leq a_1 \log \frac{a_1}{b_1} + a_2 \log \frac{a_2}{b_2}.$$

KL joint convexity: by applying log-sum inequality pointwise and summing over x ,

$$\text{KL}(\lambda p_1 + (1 - \lambda) p_2 \| \lambda q_1 + (1 - \lambda) q_2) \leq \lambda \text{KL}(p_1 \| q_1) + (1 - \lambda) \text{KL}(p_2 \| q_2).$$

Gaussian density: $\text{1D } x \sim \mathcal{N}(\mu, \sigma^2)$, $p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$. $n\text{D}$

$x \sim \mathcal{N}(\mu, \Sigma)$, $p(x) = \frac{\exp[-\frac{1}{2}(x-\mu)^{\top} \Sigma^{-1}(x-\mu)]}{(2\pi)^{n/2} |\Sigma|^{1/2}}$. **Gaussian KL**: for $\mathcal{N}_0 = \mathcal{N}(\mu_0, \Sigma_0)$ and $\mathcal{N}_1 = \mathcal{N}(\mu_1, \Sigma_1)$ in $\mathbb{R}^d$,

$$\text{KL}(\mathcal{N}_0 \| \mathcal{N}_1) = \frac{1}{2} \left(\text{tr}(\Sigma_1^{-1} \Sigma_0) + (\mu_1 - \mu_0)^{\top} \Sigma_1^{-1} (\mu_1 - \mu_0) - d + \log \frac{|\Sigma_1|}{|\Sigma_0|} \right).$$

JSD metric: $\sqrt{\text{JSD}}$ satisfies triangle inequality:

$$\sqrt{\text{JSD}(p_1 \| p_2)} + \sqrt{\text{JSD}(p_2 \| p_3)} \geq \sqrt{\text{JSD}(p_1 \| p_3)}.$$

W-TV-JSD bound: assume $\|z - z'\|_2 \leq C$. Let

$$\alpha = \text{TV}(\mu, \nu) = \frac{1}{2} \|\mu - \nu\|_1, \quad \eta(z) = \min\{\mu(z), \nu(z)\}, \quad \sum_z \eta(z) = 1 - \alpha.$$

Residuals: $\mu_{\text{res}} = \mu - \eta$, $\nu_{\text{res}} = \nu - \eta$, $\tilde{\mu} = \mu_{\text{res}}/\alpha$, $\tilde{\nu} = \nu_{\text{res}}/\alpha$ if $\alpha > 0$. Coupling:

$$\gamma(z, z') = \eta(z) \mathbf{1}_{\{z=z'\}} + \alpha \tilde{\mu}(z) \tilde{\nu}(z').$$

Its marginals are μ, ν , so

$$W(\mu, \nu) \leq \sum_{z, z'} \gamma(z, z') \|z - z'\|_2 \leq C\alpha = \text{CTV}(\mu, \nu).$$

Pinsker inequality: standard form under natural logarithm:

$$\text{KL}(p\|q) \geq 2\text{TV}(p, q)^2, \quad \text{TV}(p, q) \leq \sqrt{\frac{1}{2} \text{KL}(p\|q)}.$$